

Systems of Simultaneous Linear Differential Equations

Differential equations involving **two or more dependent variables** and **one independent variable** is called simultaneous differential equations.

Dependent variables $\rightarrow x, y$

Independent variable $\rightarrow t$

$$D = \frac{d}{dt}$$

$$f_1(D)x + f_2(D)y = F(t)$$

$$g_1(D)x + g_2(D)y = G(t)$$

$$Dx + y = e^t$$

$$Dy - x = e^{-t}$$

dependent variables



independent variable



Simultaneous differential equations can be solved by the following methods

- **Method of elimination(using D operator)**
- **Method of differentiation**

We consider first order simultaneous differential equations with constant coefficients only.

Given system can be solved as follows:

- 1. Eliminate from the given system of equations resulting a differential equation exclusively in x .**
- 2. Solve the differential equation in x by usual methods to obtain x as a function of t .**
- 3. Substitute value of x and its derivatives in one of the simultaneous equations to get an equation in y .**
- 4. Solve for y by usual methods to obtain its value as a function of t .**

Given system of equations can be solved as follows

- Eliminate y from the given system of equations in order to get a differential equation involving x only.
- Solve the equation for x as a function of t .
- Substitute value of x and its derivatives in one of the simultaneous equations to get an equation in y .
- Solve this equation for y in terms of t .

Example 1

$$\frac{dx}{dt} - y = t$$

$$\frac{dy}{dt} + x = 1$$

$$D = \frac{d}{dt}$$

$$Dx - y = t \longrightarrow (1)$$

$$x + Dy = 1 \longrightarrow (2)$$

$$\text{Differentiating (1) w.r.t } t \quad D^2x - Dy = 1 \longrightarrow (3)$$

$$(2) + (3) \quad (D^2 + 1)x = 2$$

$$x = e^{mt}$$

Auxiliary equation $(m^2 + 1) = 0 \quad m = \pm i$

Complementary equation $C.F = A \cos t + B \sin t$

Particular integral $P.I = \frac{1}{1 + D^2} 2 = (1 - D^2 + \dots) 2 = 2$

$$x = A \cos t + B \sin t + 2$$

$$\begin{aligned} y = Dx - t &= D(A \cos t + B \sin t + 2) - t \\ &= -A \sin t + B \cos t - t \end{aligned}$$

Example 2

$$Dx + y = e^t$$



(1)

$$Dy - x = e^{-t}$$



(2)

$$D = \frac{d}{dt}$$

Multiplying (1) by D

$$D^2 x + Dy = e^t \longrightarrow (3)$$

$$(3) - (2) \quad (D^2 + 1)x = e^t - e^{-t}$$

$$x = e^{mt}$$

Auxiliary equation

$$(m^2 + 1) = 0 \quad m = \pm i$$

Complementary equation $C.F = A \cos t + B \sin t$

Particular integral

$$P.I = \frac{1}{1 + D^2} e^t - \frac{1}{1 + D^2} e^{-t}$$

$$\begin{aligned}
 P.I &= \frac{1}{1+1^2} e^t - \frac{1}{1+(-1)^2} e^{-t} \\
 &= \frac{1}{2} e^t - \frac{1}{2} e^{-t}
 \end{aligned}$$

$$x = A \cos t + B \sin t + \frac{1}{2} e^t - \frac{1}{2} e^{-t}$$

From (1)

$$y = e^t - Dx$$

$$y = e^t - D \left(A \cos t + B \sin t + \frac{1}{2} e^t - \frac{1}{2} e^{-t} \right)$$

$$Y = e^t + A \sin t - B \cos t - \frac{1}{2} e^t - \frac{1}{2} e^{-t}$$

Example 3

$$Dx + y = \sin t \longrightarrow (1)$$

$$Dy + x = \cos t \longrightarrow (2)$$

Multiplying (1) by D $D^2x + Dy = \cos t \longrightarrow (3)$

$$(3) - (2) \quad (D^2 - 1)x = \cos t - \cos t = 0$$

$$(D^2 - 1)x = 0$$

$$x = e^{mt} \quad \text{Auxiliary equation} \quad (m^2 - 1) = 0 \quad m = \pm 1$$

$$\text{Complementary equation} \quad C.F = Ae^t + Be^{-t}$$

No particular integral

$$x = Ae^t + Be^{-t}$$

From Eq (1)

$$y = \sin t - Dx$$

$$= \sin t - \frac{d}{dt}(Ae^t + Be^{-t})$$

$$= \sin t - Ae^t + Be^{-t}$$

$$x = Ae^t + Be^{-t}$$

$$y = \sin t - Ae^t + Be^{-t}$$

3rd Method:

$$(2D - 4)x + (D - 1)y = e^t \longrightarrow (1)$$

$$(D + 3)x + y = 0 \longrightarrow (2)$$

$$\begin{pmatrix} (2D - 4) & (D - 1) \\ (D + 3) & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e^t \\ 0 \end{pmatrix}$$

Note:

$$\begin{array}{l} ax + by = c \\ px + qy = r \end{array} \quad \begin{pmatrix} a & b \\ p & q \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} c \\ r \end{pmatrix}$$

$$x = \frac{\begin{vmatrix} c & b \\ r & q \end{vmatrix}}{\begin{vmatrix} a & b \\ p & q \end{vmatrix}}$$

$$y = \frac{\begin{vmatrix} a & c \\ p & r \end{vmatrix}}{\begin{vmatrix} a & b \\ p & q \end{vmatrix}}$$

$$x = \frac{\begin{vmatrix} e^t & (D-1) \\ 0 & 1 \end{vmatrix}}{\begin{vmatrix} (2D-4) & (D-1) \\ (D+3) & 1 \end{vmatrix}}$$

$$x = \frac{e^t}{(2D-4) - (D-1)(D+3)}$$

$$x = \frac{e^t}{2D-4-D^2-2D+3}$$

$$x = \frac{1}{-D^2-1} e^t$$

$$x = -\frac{1}{D^2+1} e^t$$

$$x = -\frac{1}{2} e^t + A_1 \cos(t) + B_1 \sin(t)$$

$$y = \frac{\begin{vmatrix} 2D - 4 & e^t \\ D + 3 & 0 \end{vmatrix}}{\begin{vmatrix} 2D - 4 & (D - 1) \\ D + 3 & 1 \end{vmatrix}}$$

$$y = -\frac{(D + 3)e^t}{2D - 4 - (D^2 + 2D - 3)}$$

$$y = \frac{e^t + 3e^t}{-(D^2 + 1)}$$

$$y = -\frac{4e^t}{(D^2 + 1)}$$

$$y = -2e^t + A_3 \cos(t) + A_4 \sin(t)$$